



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0618 – DESIGN AND ANALYSIS OF ALGORITHM

EXPERIMENT NO: 01

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Experiment 1: Linear Search (Iterative)

Program:

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i] == key:  
            return i  
    return -1  
  
arr = [10, 25, 30, 45, 50]  
key = 30  
result = linear_search(arr, key)  
if result != -1:  
    print("Key found at index", result)  
else:  
    print("Key not found")
```

Output:

main.py	Visualize	Run ▶	Output
<pre>1 def linear_search(arr, key): 2 for i in range(len(arr)): 3 if arr[i] == key: 4 return i 5 return -1 6 arr = [10, 25, 30, 45, 50] 7 key = 30 8 result = linear_search(arr, key) 9 if result != -1: 10 print("Key found at index", result) 11 else: 12 print("Key not found")</pre>			Key found at index 2

Experiment 2: Binary Search (Recursive)

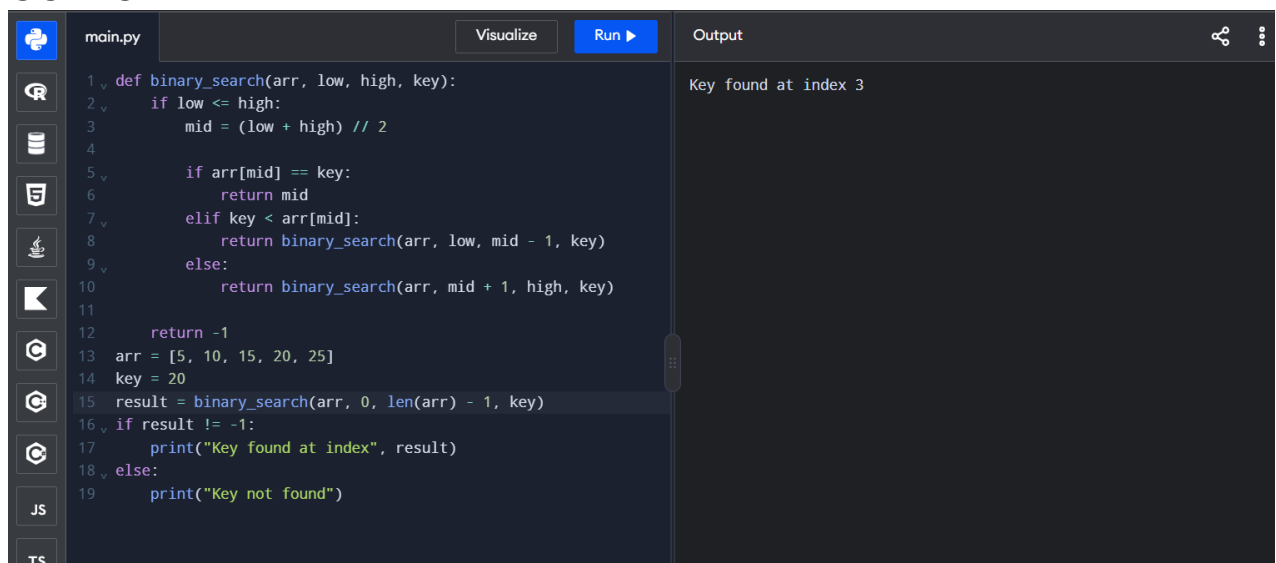
Program:

```
def binary(arr, low, high, key):
    if low <= high:
        mid = (low + high) // 2
        if arr[mid] == key:
            return mid
        elif key < arr[mid]:
            return binary(arr, low, mid - 1, key)
        else:
            return binary(arr, mid + 1, high, key)

    return -1

arr = [5, 10, 15, 20, 25]
key = 20
result = binary(arr, 0, len(arr) - 1, key)
if result != -1:
    print("Key found at index", result)
else:
    print("Key not found")
```

OUTPUT



```
main.py Visualize Run ▶ Output
1 def binary_search(arr, low, high, key):
2     if low <= high:
3         mid = (low + high) // 2
4
5         if arr[mid] == key:
6             return mid
7         elif key < arr[mid]:
8             return binary_search(arr, low, mid - 1, key)
9         else:
10            return binary_search(arr, mid + 1, high, key)
11
12    return -1
13 arr = [5, 10, 15, 20, 25]
14 key = 20
15 result = binary_search(arr, 0, len(arr) - 1, key)
16 if result != -1:
17     print("Key found at index", result)
18 else:
19     print("Key not found")
```

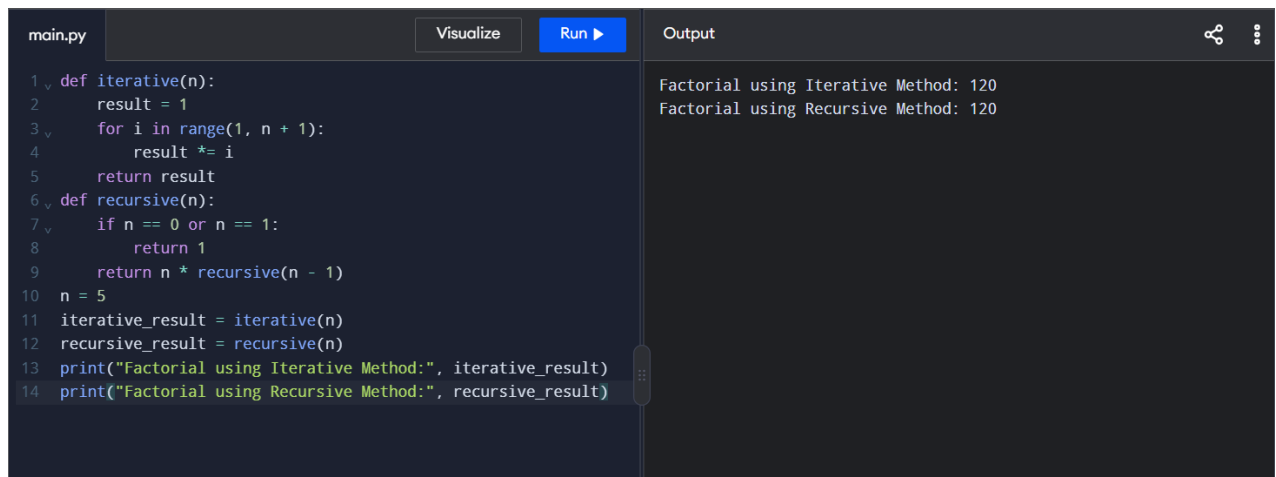
Key found at index 3

Experiment 3: Factorial (Iterative vs Recursive)

Program:

```
def iterative(n):  
    result = 1  
    for i in range(1, n + 1):  
        result *= i  
    return result  
  
def recursive(n):  
    if n == 0 or n == 1:  
        return 1  
    return n * recursive(n - 1)  
  
n = 5  
  
iterative_result = iterative(n)  
recursive_result = recursive(n)  
  
print("Factorial using Iterative Method:", iterative_result)  
print("Factorial using Recursive Method:", recursive_result)
```

OUTPUT



The screenshot shows a code editor with a dark theme. The left pane displays the Python code for calculating factorials using both iterative and recursive methods. The code defines two functions, `iterative(n)` and `recursive(n)`, and then calculates the factorial of 5 using both methods. The right pane, titled "Output", shows the results of the program execution: "Factorial using Iterative Method: 120" and "Factorial using Recursive Method: 120". The code editor has a tab labeled "main.py" and buttons for "Visualize" and "Run".

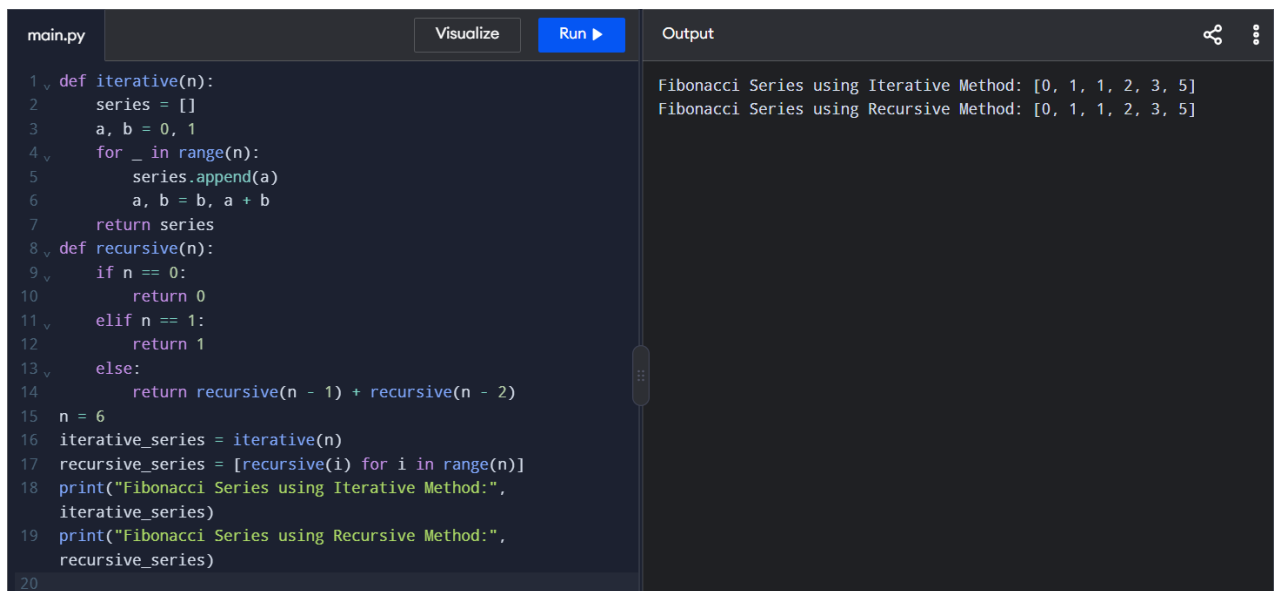
```
main.py Visualize Run ▶ Output  
1 def iterative(n):  
2     result = 1  
3     for i in range(1, n + 1):  
4         result *= i  
5     return result  
6 def recursive(n):  
7     if n == 0 or n == 1:  
8         return 1  
9     return n * recursive(n - 1)  
10 n = 5  
11 iterative_result = iterative(n)  
12 recursive_result = recursive(n)  
13 print("Factorial using Iterative Method:", iterative_result)  
14 print("Factorial using Recursive Method:", recursive_result)  
  
Factorial using Iterative Method: 120  
Factorial using Recursive Method: 120
```

Experiment 4: Fibonacci Series

Program:

```
def iterative(n):  
    series = []  
    a, b = 0, 1  
    for _ in range(n):  
        series.append(a)  
        a, b = b, a + b  
    return series  
  
def recursive(n):  
    if n == 0:  
        return 0  
    elif n == 1:  
        return 1  
    else:  
        return recursive(n - 1) + recursive(n - 2)  
  
n = 6  
  
iterative_series = iterative(n)  
  
recursive_series = [recursive(i) for i in range(n)]  
  
print("Fibonacci Series using Iterative Method:", iterative_series)  
  
print("Fibonacci Series using Recursive Method:", recursive_series)
```

OUTPUT:



The screenshot shows a code editor with a file named 'main.py'. The code defines two functions: 'iterative(n)' which uses a loop to build a list of Fibonacci numbers, and 'recursive(n)' which uses recursion to calculate the nth Fibonacci number. The main part of the code sets n=6, generates the series using both methods, and prints them. The output panel on the right shows the results: 'Fibonacci Series using Iterative Method: [0, 1, 1, 2, 3, 5]' and 'Fibonacci Series using Recursive Method: [0, 1, 1, 2, 3, 5]'. The code is written in Python with syntax highlighting, and the output is displayed in a dark-themed window.

```
main.py Visualize Run ▶ Output  
1 def iterative(n):  
2     series = []  
3     a, b = 0, 1  
4     for _ in range(n):  
5         series.append(a)  
6         a, b = b, a + b  
7     return series  
8 def recursive(n):  
9     if n == 0:  
10        return 0  
11    elif n == 1:  
12        return 1  
13    else:  
14        return recursive(n - 1) + recursive(n - 2)  
15  
16 n = 6  
17 iterative_series = iterative(n)  
18 recursive_series = [recursive(i) for i in range(n)]  
19 print("Fibonacci Series using Iterative Method:",  
20     iterative_series)  
21 print("Fibonacci Series using Recursive Method:",  
22     recursive_series)  
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```

Experiment 5: Matrix Multiplication

Program:

```
A = [[1, 2],
      [3, 4]]
B = [[5, 6],
      [7, 8]]
result = [[0, 0],
          [0, 0]]
for i in range(len(A)):
    for j in range(len(B[0])):
        for k in range(len(B)):
            result[i][j] += A[i][k] * B[k][j]
print("Resultant Matrix:")
print(result)
```

OUTPUT

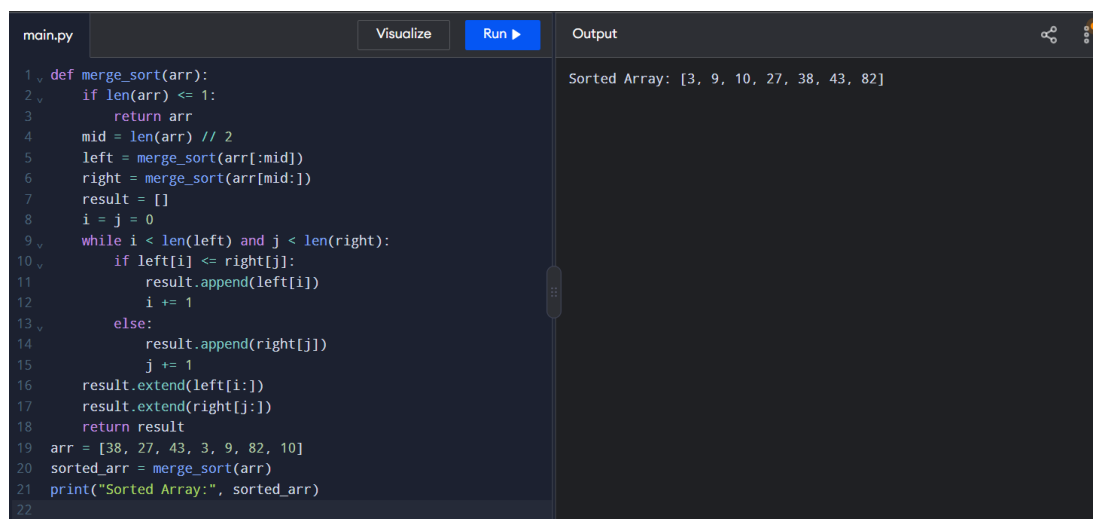
main.py	Visualize	Run ▶	Output
<pre>1 A = [[1, 2], 2 [3, 4]] 3 B = [[5, 6], 4 [7, 8]] 5 result = [[0, 0], 6 [0, 0]] 7 for i in range(len(A)): 8 for j in range(len(B[0])): 9 for k in range(len(B)): 10 result[i][j] += A[i][k] * B[k][j] 11 print("Resultant Matrix:") 12 print(result) 13</pre>			<pre>Resultant Matrix: [[19, 22], [43, 50]]</pre>

Experiment 6: Merge Sort

Program:

```
def merge_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    mid = len(arr) // 2  
  
    left = merge_sort(arr[:mid])  
    right = merge_sort(arr[mid:])  
  
    result = []  
    i = j = 0  
  
    while i < len(left) and j < len(right):  
        if left[i] <= right[j]:  
            result.append(left[i])  
            i += 1  
        else:  
            result.append(right[j])  
            j += 1  
    result.extend(left[i:])  
    result.extend(right[j:])  
  
    return result  
  
arr = [38, 27, 43, 3, 9, 82, 10]  
sorted_arr = merge_sort(arr)  
print("Sorted Array:", sorted_arr)
```

OUTPUT



The screenshot shows a code editor with a dark theme. The left pane displays the Python code for the Merge Sort algorithm, with line numbers 1 through 22. The right pane, titled 'Output', shows the result of running the code: 'Sorted Array: [3, 9, 10, 27, 38, 43, 82]'. The code is as follows:

```
main.py Visualize Run  
1 def merge_sort(arr):  
2     if len(arr) <= 1:  
3         return arr  
4     mid = len(arr) // 2  
5     left = merge_sort(arr[:mid])  
6     right = merge_sort(arr[mid:])  
7     result = []  
8     i = j = 0  
9     while i < len(left) and j < len(right):  
10        if left[i] <= right[j]:  
11            result.append(left[i])  
12            i += 1  
13        else:  
14            result.append(right[j])  
15            j += 1  
16    result.extend(left[i:])  
17    result.extend(right[j:])  
18    return result  
19 arr = [38, 27, 43, 3, 9, 82, 10]  
20 sorted_arr = merge_sort(arr)  
21 print("Sorted Array:", sorted_arr)  
22
```

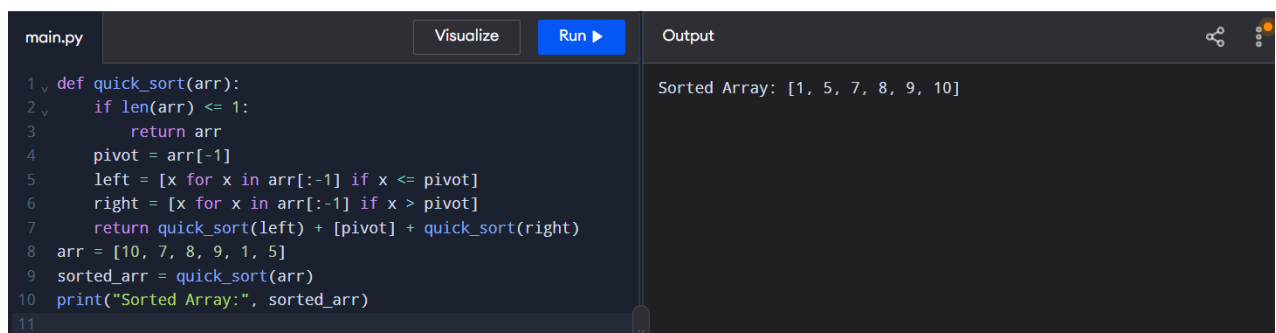
Output: Sorted Array: [3, 9, 10, 27, 38, 43, 82]

Experiment 7: Quick Sort

Program:

```
def quick_sort(arr):  
    if len(arr) <= 1:  
        return arr  
  
    pivot = arr[-1]  
    left = [x for x in arr[:-1] if x <= pivot]  
    right = [x for x in arr[:-1] if x > pivot]  
    return quick_sort(left) + [pivot] + quick_sort(right)  
  
arr = [10, 7, 8, 9, 1, 5]  
sorted_arr = quick_sort(arr)  
print("Sorted Array:", sorted_arr)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code is a Python implementation of the Quick Sort algorithm. It defines a function 'quick_sort' that takes an array 'arr' and returns a sorted array. The function uses a pivot element (the last element of the array) and partitions the array into two sub-arrays: 'left' (elements less than or equal to the pivot) and 'right' (elements greater than the pivot). The function then recursively sorts these sub-arrays and concatenates the results. The main part of the program initializes an array 'arr' with the values [10, 7, 8, 9, 1, 5], calls 'quick_sort(arr)' to get 'sorted_arr', and prints the result. The output window on the right shows the printed result: 'Sorted Array: [1, 5, 7, 8, 9, 10]'.

```
main.py Visualize Run ▶ Output  
1 def quick_sort(arr):  
2     if len(arr) <= 1:  
3         return arr  
4     pivot = arr[-1]  
5     left = [x for x in arr[:-1] if x <= pivot]  
6     right = [x for x in arr[:-1] if x > pivot]  
7     return quick_sort(left) + [pivot] + quick_sort(right)  
8 arr = [10, 7, 8, 9, 1, 5]  
9 sorted_arr = quick_sort(arr)  
10 print("Sorted Array:", sorted_arr)  
11
```

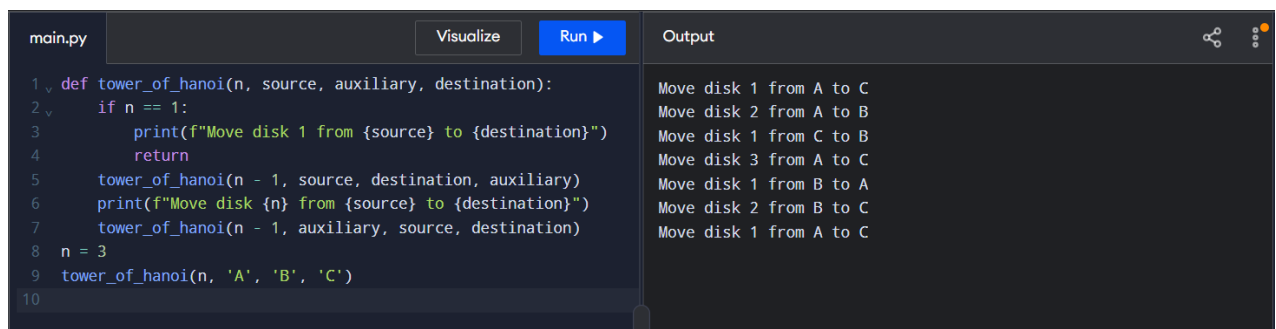
Sorted Array: [1, 5, 7, 8, 9, 10]

Experiment 8: Tower of Hanoi

Program:

```
def tower_of_hanoi(n, source, auxiliary, destination):  
    if n == 1:  
        print(f'Move disk 1 from {source} to {destination}')  
        return  
    tower_of_hanoi(n - 1, source, destination, auxiliary)  
    print(f'Move disk {n} from {source} to {destination}')  
    tower_of_hanoi(n - 1, auxiliary, source, destination)  
  
n = 3  
tower_of_hanoi(n, 'A', 'B', 'C')
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code defines a recursive function 'tower_of_hanoi' and calls it with 'n=3' and towers 'A', 'B', and 'C'. The output window on the right displays the sequence of moves required to solve the puzzle.

```
main.py Visualize Run ▶ Output
```

```
1 def tower_of_hanoi(n, source, auxiliary, destination):  
2     if n == 1:  
3         print(f'Move disk 1 from {source} to {destination}')  
4         return  
5     tower_of_hanoi(n - 1, source, destination, auxiliary)  
6     print(f'Move disk {n} from {source} to {destination}')  
7     tower_of_hanoi(n - 1, auxiliary, source, destination)  
8     n = 3  
9     tower_of_hanoi(n, 'A', 'B', 'C')
```

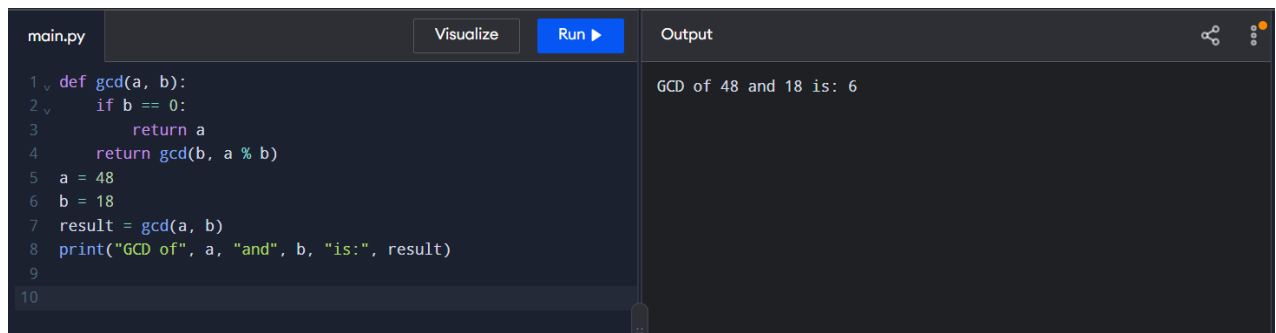
```
Move disk 1 from A to C  
Move disk 2 from A to B  
Move disk 1 from C to B  
Move disk 3 from A to C  
Move disk 1 from B to A  
Move disk 2 from B to C  
Move disk 1 from A to C
```

Experiment 9: GCD (Euclidean Algorithm)

Program:

```
def gcd(a, b):  
    if b == 0:  
        return a  
    return gcd(b, a % b)  
  
a = 48  
b = 18  
  
result = gcd(a, b)  
print("GCD of", a, "and", b, "is:", result)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code is a Python program that implements the Euclidean algorithm to find the GCD of 48 and 18. The code is as follows:

```
1 def gcd(a, b):  
2     if b == 0:  
3         return a  
4     return gcd(b, a % b)  
5  
6 a = 48  
7 b = 18  
8  
9 result = gcd(a, b)  
10 print("GCD of", a, "and", b, "is:", result)
```

There are 'Visualize' and 'Run' buttons above the code. The 'Run' button is highlighted. To the right of the code editor is an 'Output' panel that displays the result of the program execution:

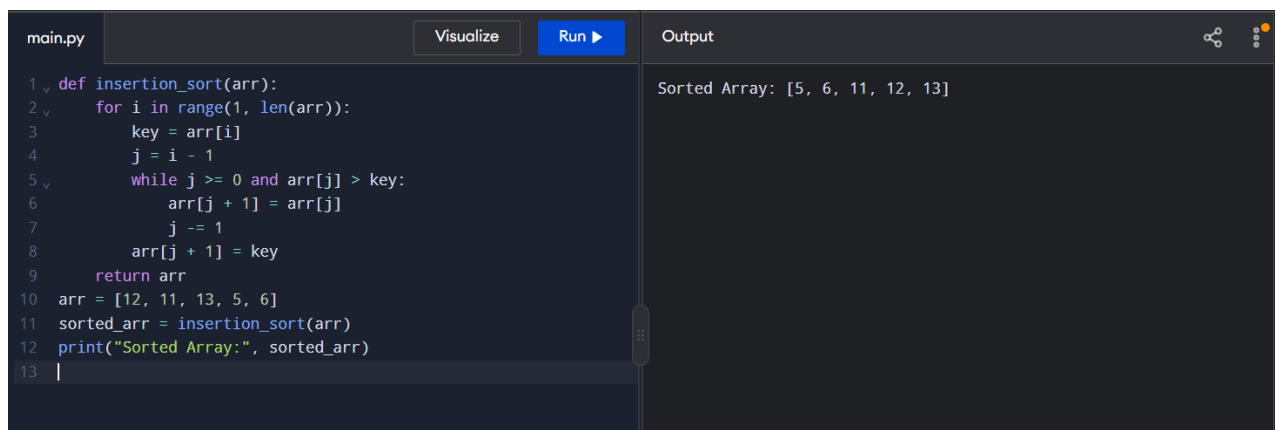
```
GCD of 48 and 18 is: 6
```

Experiment 10: Insertion Sort

Program:

```
def insertion_sort(arr):  
    for i in range(1, len(arr)):  
        key = arr[i]  
        j = i - 1  
        while j >= 0 and arr[j] > key:  
            arr[j + 1] = arr[j]  
            j -= 1  
        arr[j + 1] = key  
    return arr  
  
arr = [12, 11, 13, 5, 6]  
sorted_arr = insertion_sort(arr)  
print("Sorted Array:", sorted_arr)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code implements the Insertion Sort algorithm. The array [12, 11, 13, 5, 6] is sorted in-place, and the sorted array [5, 6, 11, 12, 13] is printed. The output window on the right displays the message 'Sorted Array: [5, 6, 11, 12, 13]'.

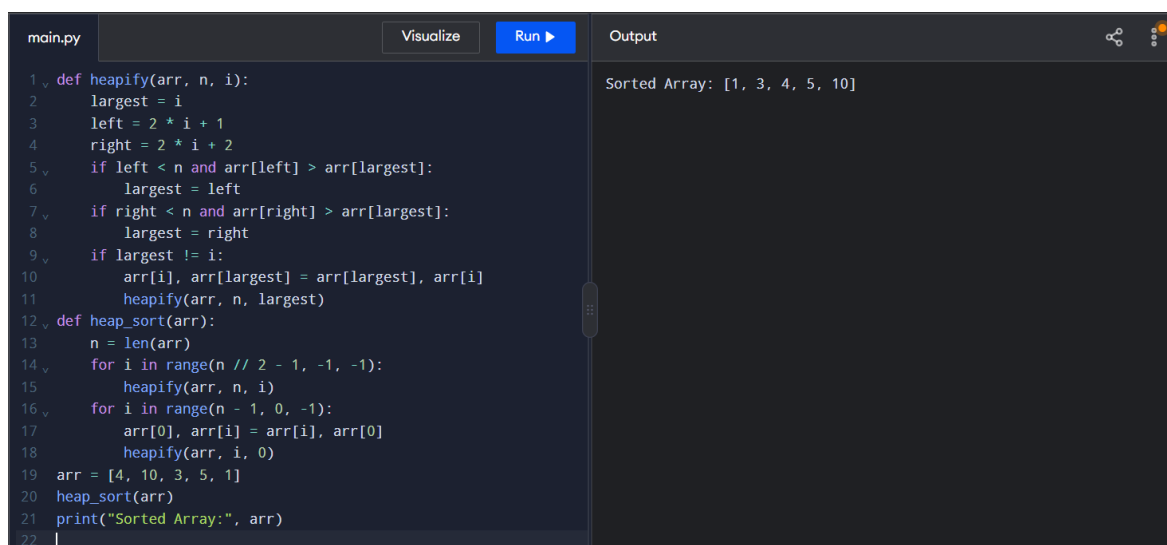
```
main.py Visualize Run ▶  
1 def insertion_sort(arr):  
2     for i in range(1, len(arr)):  
3         key = arr[i]  
4         j = i - 1  
5         while j >= 0 and arr[j] > key:  
6             arr[j + 1] = arr[j]  
7             j -= 1  
8         arr[j + 1] = key  
9     return arr  
10 arr = [12, 11, 13, 5, 6]  
11 sorted_arr = insertion_sort(arr)  
12 print("Sorted Array:", sorted_arr)  
13 |  
  
Output  
Sorted Array: [5, 6, 11, 12, 13]
```

Experiment 11: Heap Sort

Program:

```
def heapify(arr, n, i):  
    largest = i  
    left = 2 * i + 1  
    right = 2 * i + 2  
    if left < n and arr[left] > arr[largest]:  
        largest = left  
    if right < n and arr[right] > arr[largest]:  
        largest = right  
    if largest != i:  
        arr[i], arr[largest] = arr[largest], arr[i]  
        heapify(arr, n, largest)  
  
def heap_sort(arr):  
    n = len(arr)  
    for i in range(n // 2 - 1, -1, -1):  
        heapify(arr, n, i)  
    for i in range(n - 1, 0, -1):  
        arr[0], arr[i] = arr[i], arr[0]  
        heapify(arr, i, 0)  
  
arr = [4, 10, 3, 5, 1]  
heap_sort(arr)  
print("Sorted Array:", arr)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code is a Python implementation of heap sort. It defines two functions: 'heapify' and 'heap_sort'. The 'heapify' function takes an array 'arr', its size 'n', and an index 'i'. It finds the largest element among 'i' and its children, and swaps it with 'i' if necessary. The 'heap_sort' function uses 'heapify' to sort the array in descending order. The array is initialized as [4, 10, 3, 5, 1]. The output of the program is displayed in a separate pane on the right, showing 'Sorted Array: [1, 3, 4, 5, 10]'. The code editor has a dark theme and includes buttons for 'Visualize' and 'Run'.

```
main.py Visualize Run ▶ Output  
1 def heapify(arr, n, i):  
2     largest = i  
3     left = 2 * i + 1  
4     right = 2 * i + 2  
5     if left < n and arr[left] > arr[largest]:  
6         largest = left  
7     if right < n and arr[right] > arr[largest]:  
8         largest = right  
9     if largest != i:  
10        arr[i], arr[largest] = arr[largest], arr[i]  
11        heapify(arr, n, largest)  
12 def heap_sort(arr):  
13     n = len(arr)  
14     for i in range(n // 2 - 1, -1, -1):  
15         heapify(arr, n, i)  
16     for i in range(n - 1, 0, -1):  
17         arr[0], arr[i] = arr[i], arr[0]  
18         heapify(arr, i, 0)  
19 arr = [4, 10, 3, 5, 1]  
20 heap_sort(arr)  
21 print("Sorted Array:", arr)  
22 |  
Sorted Array: [1, 3, 4, 5, 10]
```

Experiment 12: Counting Inversions

Program:

```
def merge_and_count(arr, left, mid, right):  
    i = left  
    j = mid + 1  
    temp = []  
    inversions = 0  
    while i <= mid and j <= right:  
        if arr[i] <= arr[j]:  
            temp.append(arr[i])  
            i += 1  
        else:  
            temp.append(arr[j])  
            inversions += (mid - i + 1)  
            j += 1  
    while i <= mid:  
        temp.append(arr[i])  
        i += 1  
    while j <= right:  
        temp.append(arr[j])  
        j += 1  
    for k in range(len(temp)):  
        arr[left + k] = temp[k]  
    return inversions  
  
def count_inversions(arr, left, right):  
    if left >= right:  
        return 0  
    mid = (left + right) // 2  
    inversions = 0  
    inversions += count_inversions(arr, left, mid)  
    inversions += count_inversions(arr, mid + 1, right)
```

```
    inversions += merge_and_count(arr, left, mid, right)

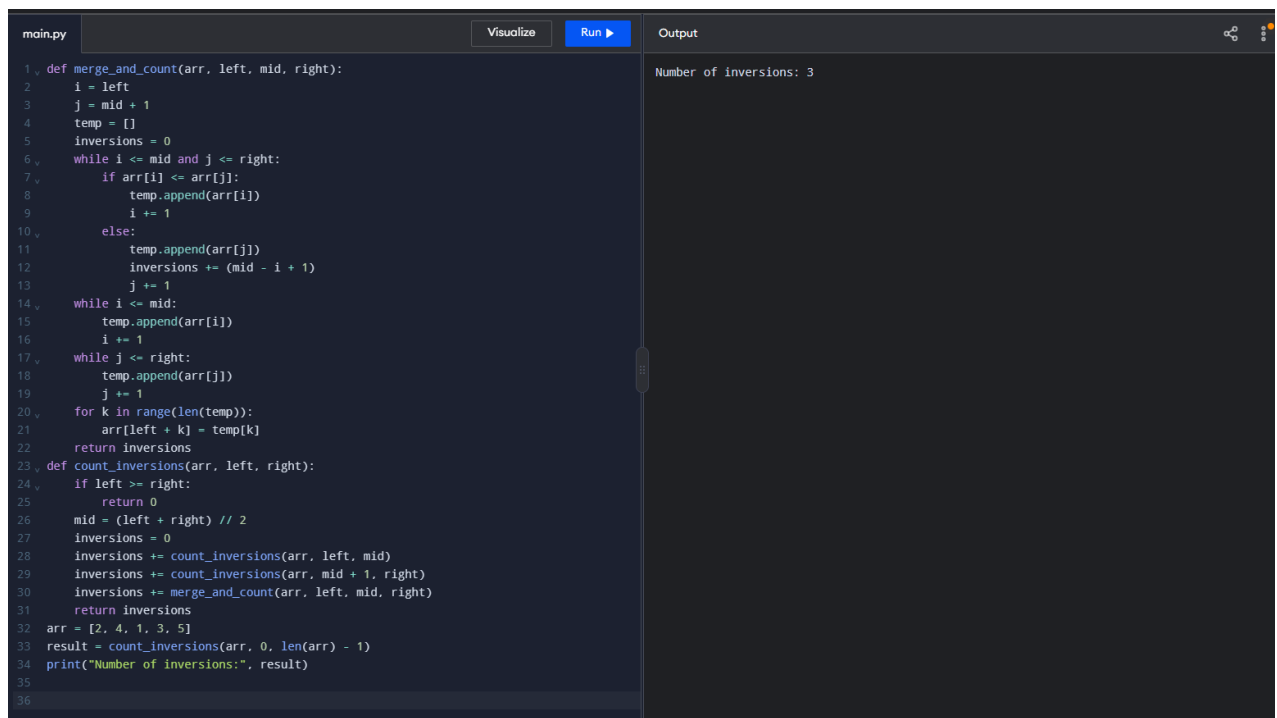
    return inversions

arr = [2, 4, 1, 3, 5]

result = count_inversions(arr, 0, len(arr) - 1)

print("Number of inversions:", result)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code implements a recursive function to count inversions in an array. The array is [2, 4, 1, 3, 5]. The output of the program is 'Number of inversions: 3'.

```
main.py Visualize Run ▶ Output
1, def merge_and_count(arr, left, mid, right):
2,     i = left
3,     j = mid + 1
4,     temp = []
5,     inversions = 0
6,     while i <= mid and j <= right:
7,         if arr[i] <= arr[j]:
8,             temp.append(arr[i])
9,             i += 1
10,        else:
11,            temp.append(arr[j])
12,            inversions += (mid - i + 1)
13,            j += 1
14,        while i <= mid:
15,            temp.append(arr[i])
16,            i += 1
17,        while j <= right:
18,            temp.append(arr[j])
19,            j += 1
20,        for k in range(len(temp)):
21,            arr[left + k] = temp[k]
22,        return inversions
23, def count_inversions(arr, left, right):
24,     if left >= right:
25,         return 0
26,     mid = (left + right) // 2
27,     inversions = 0
28,     inversions += count_inversions(arr, left, mid)
29,     inversions += count_inversions(arr, mid + 1, right)
30,     inversions += merge_and_count(arr, left, mid, right)
31,     return inversions
32, arr = [2, 4, 1, 3, 5]
33, result = count_inversions(arr, 0, len(arr) - 1)
34, print("Number of inversions:", result)
35,
36,
```

Number of inversions: 3

Experiment 13: Maximum Subarray Sum

Program:

```
def kadane(arr):
    current_sum = arr[0]
    max_sum = arr[0]
    for i in range(1, len(arr)):
        current_sum = max(arr[i], current_sum + arr[i])
        max_sum = max(max_sum, current_sum)
    return max_sum

def max_crossing_sum(arr, low, mid, high):
    left_sum = float('-inf')
    total = 0
    for i in range(mid, low - 1, -1):
        total += arr[i]
        left_sum = max(left_sum, total)
    right_sum = float('-inf')
    total = 0
    for i in range(mid + 1, high + 1):
        total += arr[i]
        right_sum = max(right_sum, total)
    return left_sum + right_sum

def divide_conquer(arr, low, high):
    if low == high:
        return arr[low]
    mid = (low + high) // 2
    left = divide_conquer(arr, low, mid)
    right = divide_conquer(arr, mid + 1, high)
    cross = max_crossing_sum(arr, low, mid, high)
    return max(left, right, cross)

arr = [-2, -3, 4, -1, -2, 1, 5, -3]
```

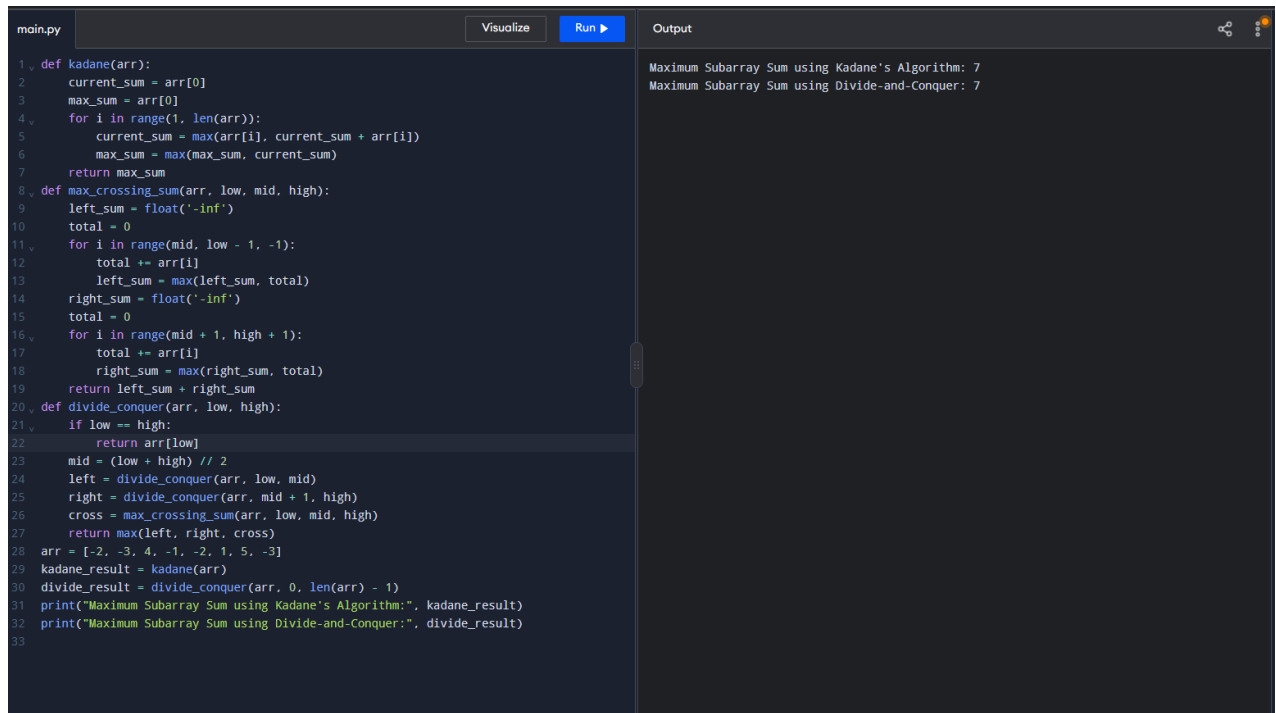
```
kadane_result = kadane(arr)
```

```
divide_result = divide_conquer(arr, 0, len(arr) - 1)
```

```
print("Maximum Subarray Sum using Kadane's Algorithm:", kadane_result)
```

```
print("Maximum Subarray Sum using Divide-and-Conquer:", divide_result)
```

OUTPUT



The screenshot shows a code editor with a file named `main.py`. The code implements two algorithms to find the maximum subarray sum. The left pane shows the code, and the right pane shows the output.

```
1 def kadane(arr):
2     current_sum = arr[0]
3     max_sum = arr[0]
4     for i in range(1, len(arr)):
5         current_sum = max(arr[i], current_sum + arr[i])
6         max_sum = max(max_sum, current_sum)
7     return max_sum
8
9 def max_crossing_sum(arr, low, mid, high):
10     left_sum = float('-inf')
11     total = 0
12     for i in range(mid, low - 1, -1):
13         total += arr[i]
14         left_sum = max(left_sum, total)
15     right_sum = float('-inf')
16     total = 0
17     for i in range(mid + 1, high + 1):
18         total += arr[i]
19         right_sum = max(right_sum, total)
20     return left_sum + right_sum
21
22 def divide_conquer(arr, low, high):
23     if low == high:
24         return arr[low]
25     mid = (low + high) // 2
26     left = divide_conquer(arr, low, mid)
27     right = divide_conquer(arr, mid + 1, high)
28     cross = max_crossing_sum(arr, low, mid, high)
29     return max(left, right, cross)
30
31 arr = [-2, -3, 4, -1, -2, 1, 5, -3]
32 kadane_result = kadane(arr)
33 divide_result = divide_conquer(arr, 0, len(arr) - 1)
34 print("Maximum Subarray Sum using Kadane's Algorithm:", kadane_result)
35 print("Maximum Subarray Sum using Divide-and-Conquer:", divide_result)
```

The output pane shows the following results:

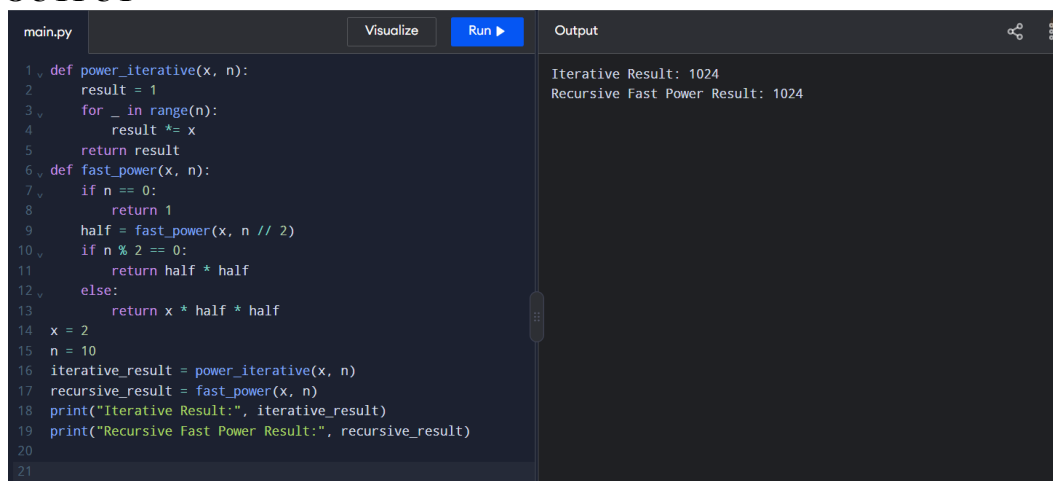
```
Maximum Subarray Sum using Kadane's Algorithm: 7
Maximum Subarray Sum using Divide-and-Conquer: 7
```

Experiment 14: Exponentiation

Program:

```
def power_iterative(x, n):  
    result = 1  
    for _ in range(n):  
        result *= x  
    return result  
  
def fast_power(x, n):  
    if n == 0:  
        return 1  
    half = fast_power(x, n // 2)  
    if n % 2 == 0:  
        return half * half  
    else:  
        return x * half * half  
  
x = 2  
n = 10  
  
iterative_result = power_iterative(x, n)  
recursive_result = fast_power(x, n)  
print("Iterative Result:", iterative_result)  
print("Recursive Fast Power Result:", recursive_result)
```

OUTPUT



The screenshot shows a code editor with a dark theme. The left pane displays the Python code from the previous block, with line numbers 1 through 21. The right pane, titled 'Output', shows the results of running the code: 'Iterative Result: 1024' and 'Recursive Fast Power Result: 1024'. Above the code editor, there are buttons for 'Visualize' and 'Run' (with a play icon). The file name 'main.py' is visible in the top left of the editor area.

```
main.py Visualize Run ▶ Output  
1 def power_iterative(x, n):  
2     result = 1  
3     for _ in range(n):  
4         result *= x  
5     return result  
6 def fast_power(x, n):  
7     if n == 0:  
8         return 1  
9     half = fast_power(x, n // 2)  
10    if n % 2 == 0:  
11        return half * half  
12    else:  
13        return x * half * half  
14 x = 2  
15 n = 10  
16 iterative_result = power_iterative(x, n)  
17 recursive_result = fast_power(x, n)  
18 print("Iterative Result:", iterative_result)  
19 print("Recursive Fast Power Result:", recursive_result)  
20  
21  
Iterative Result: 1024  
Recursive Fast Power Result: 1024
```

Experiment 15: Closest Pair of Points

Program:

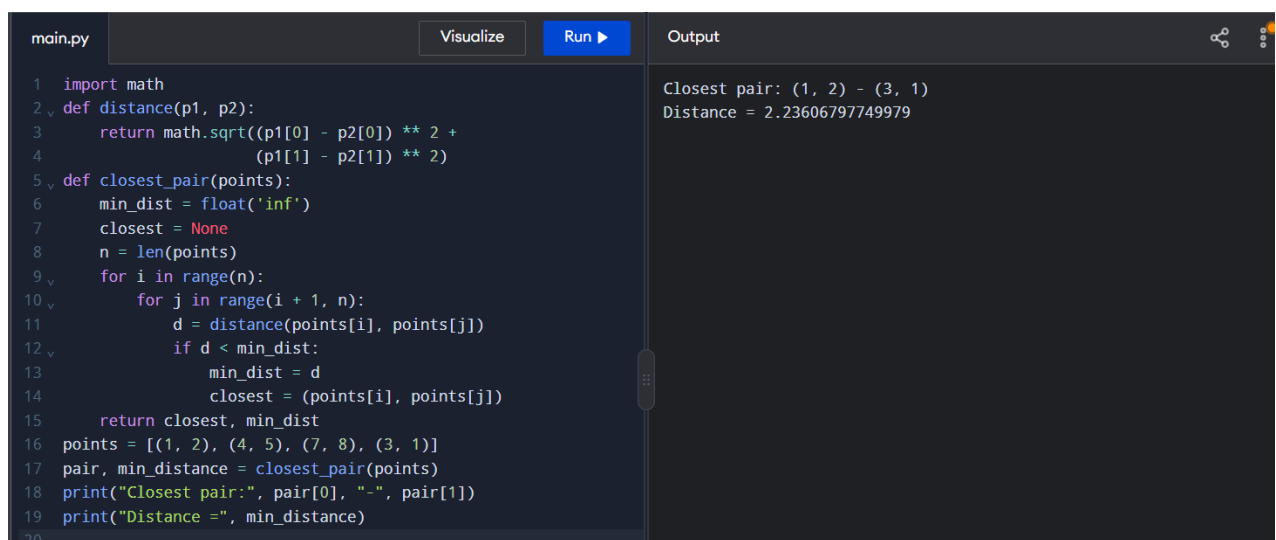
```
import math

def distance(p1, p2):
    return math.sqrt((p1[0] - p2[0]) ** 2 +
                     (p1[1] - p2[1]) ** 2)

def closest_pair(points):
    min_dist = float('inf')
    closest = None
    n = len(points)
    for i in range(n):
        for j in range(i + 1, n):
            d = distance(points[i], points[j])
            if d < min_dist:
                min_dist = d
                closest = (points[i], points[j])
    return closest, min_dist

points = [(1, 2), (4, 5), (7, 8), (3, 1)]
pair, min_distance = closest_pair(points)
print("Closest pair:", pair[0], "-", pair[1])
print("Distance =", min_distance)
```

OUTPUT



The screenshot shows a code editor with a file named 'main.py'. The code defines a 'distance' function using the Euclidean distance formula, a 'closest_pair' function that iterates through all pairs of points to find the minimum distance, and a main execution block that defines a set of points, calls 'closest_pair', and prints the results. The output panel on the right shows the execution results: 'Closest pair: (1, 2) - (3, 1)' and 'Distance = 2.23606797749979'.

```
main.py Visualize Run ▶ Output
1 import math
2 def distance(p1, p2):
3     return math.sqrt((p1[0] - p2[0]) ** 2 +
4                     (p1[1] - p2[1]) ** 2)
5 def closest_pair(points):
6     min_dist = float('inf')
7     closest = None
8     n = len(points)
9     for i in range(n):
10        for j in range(i + 1, n):
11            d = distance(points[i], points[j])
12            if d < min_dist:
13                min_dist = d
14                closest = (points[i], points[j])
15    return closest, min_dist
16 points = [(1, 2), (4, 5), (7, 8), (3, 1)]
17 pair, min_distance = closest_pair(points)
18 print("Closest pair:", pair[0], "-", pair[1])
19 print("Distance =", min_distance)
20
```

Closest pair: (1, 2) - (3, 1)
Distance = 2.23606797749979